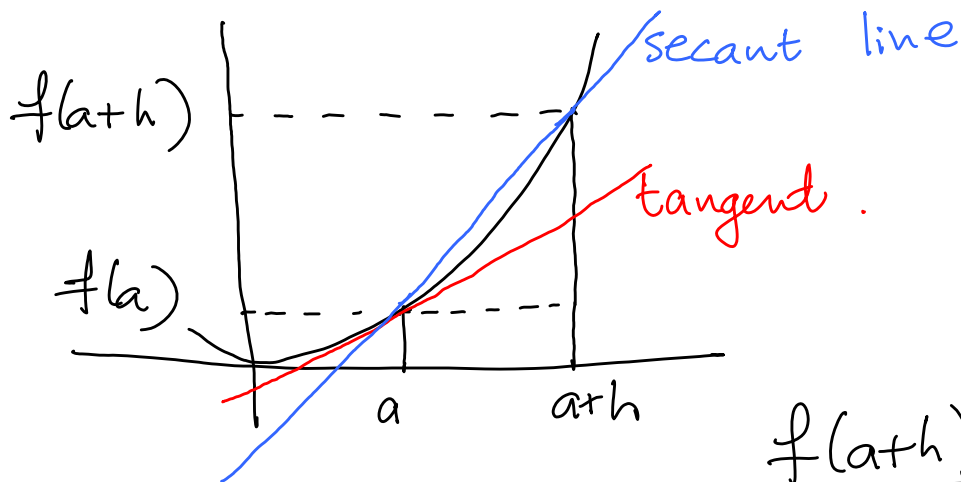


The derivative of f in a point a :



Slope of the secant line: $\frac{f(a+h) - f(a)}{h}$

Slope of the tangent in $x=a$:

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = f'(a).$$

Eg. Find from first principles $f'(3)$ if $f(x) = \frac{1}{x}$

$$f'(3) = \lim_{h \rightarrow 0} \frac{f(3+h) - f(3)}{h} \quad \left| \begin{array}{l} f(x) = \frac{1}{x} \\ \Rightarrow f'(x) = -\frac{1}{x^2} \end{array} \right.$$

$$= \lim_{h \rightarrow 0} \frac{\frac{1}{3+h} - \frac{1}{3}}{h}$$

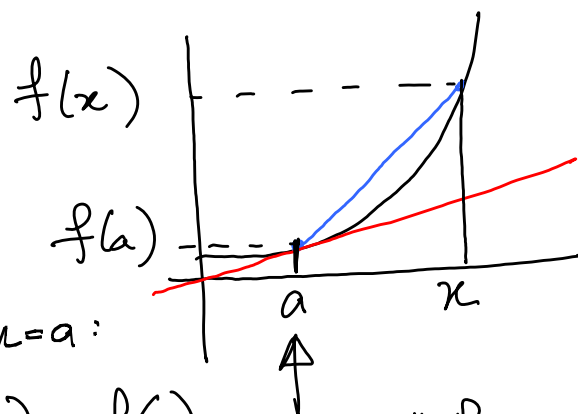
$$= \lim_{h \rightarrow 0} \frac{\frac{3 - (3+h)}{(3+h)3}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{-h}{h} \times \frac{1}{(3+h)3} = -\frac{1}{9}.$$

Alternative definition of

the derivative in a point $x=a$:

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$



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Eg $f(x) = x^2$. Find $f'(3)$.

$$f'(3) = \lim_{x \rightarrow 3} \frac{f(x) - f(3)}{x - 3}$$

$$= \lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$$

$$= \lim_{x \rightarrow 3} \frac{(x-3)(x+3)}{x-3}$$

$$= \underline{6.}$$

$$\left(\frac{0}{0} \right)$$

6.